# VITAMIN B COMPLEX

Supplement — not a cure-all

Pyridoxine neuropathy (high dose)

**CHEMICAL NAME** Combination of B vitamins: thiamine (B1), riboflavin (B2), nicotinamide (B3), pantothenic acid (B5), pyridoxine (B6), ± biotin (B7), folic acid (B9), cyanocobalamin (B12)

**CLASS** Multivitamin — water-soluble B vitamins

**TRADE NAMES** Becosules · Neurobion (B1/B6/B12) · Vitamin B Compound Strong · Polybion

## THERAPEUTIC USES / INDICATIONS

- Prevention/treatment of mixed B-vitamin deficiency: alcohol misuse, malnutrition, malabsorption, chronic illness, dialysis.
- Recovery after illness/surgery; elderly with poor intake.
- Peripheral neuropathy of nutritional origin (B1/B6/B12 combinations).
- Isoniazid-induced neuropathy prevention (pyridoxine component).

## ADMINISTRATION

- **PO** tablets/capsules/syrup with food.
- **IM/IV** combinations in some countries (inject slowly; anaphylaxis possible).

## DOSAGES

- **Vitamin B compound strong:** 1–2 tablets three times daily (alcohol misuse/treatment).
- **Maintenance:** 1–2 tablets daily.
- Doses vary by product — check individual vitamin content.

## PHARMACOKINETICS

- Water-soluble; absorbed in small intestine.
- Excess excreted in urine (riboflavin turns urine bright yellow).

## PHARMACODYNAMICS

- Coenzymes for **energy metabolism** (carbohydrate, fat, protein), nerve function, red-cell formation and DNA synthesis.

## SIDE & ADVERSE EFFECTS

### SIDE EFFECTS (COMMON)

- Bright yellow urine (riboflavin).
- Mild GI upset.
- Flushing (nicotinamide/niacin).

### ADVERSE EFFECTS (SERIOUS)

- Peripheral sensory neuropathy with long-term high-dose pyridoxine (>200 mg/day; some say >50 mg).
- Hypersensitivity (injections).
- Pyridoxine ↓ levodopa effect (without carbidopa).

## CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity.
- Levodopa without decarboxylase inhibitor (pyridoxine content).

### KEY INTERACTIONS

- **Levodopa (alone):** pyridoxine ↓ efficacy.
- **Phenytoin, phenobarbital:** high-dose pyridoxine ↓ levels.

## NURSING MANAGEMENT

- Assess nutritional status; identify specific deficiencies.
- Do not substitute for thiamine in suspected Wernicke's (needs high-dose IV thiamine).
- Review long-term need.

## PATIENT EDUCATION

- Bright yellow urine is harmless.
- A balanced diet is the best source of vitamins.
- Do not exceed recommended doses long-term.

**KEY POINTS**

- Mixed B vitamins for deficiency states.
- High-dose B6 → neuropathy.
- Yellow urine (riboflavin).
- Not a substitute for IV thiamine in Wernicke's.